

AZARIA A. REED

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PROFESSIONAL EXPERIENCE

Systems Engineer

Jan 2022 – Present

SafeHug LLC

Madison, AL

- Owned verification and validation of a production web system supporting a patented medical safety device, ensuring alignment with real-world safety intent
- Led change management and continuous monitoring, identifying risks, anomalies, and emerging issues to preserve system stability, security, and operability
- Served as the primary technical point of contact, ensuring traceability between requirements and system behavior
- Validated end-to-end system functionality, performance, and operational reliability, translating stakeholder intent into verifiable technical outcomes

Software Development Engineer in Test (SDET)

May 2026 – Present

Octave Intelligence plc • Internship

Madison, AL

- Scaled verification of a flagship industrial engineering design platform, Smart 3D, by developing an AI-driven workflow projected to reduce engineering effort across 50+ work items by more than 50% and generate production-ready test suites in an hour per class instead of 30 days
- Engineered an AI generation-and-verification asset with modular orchestration and deterministic verification for Smart 3D, evolving it through 14 production versions to generate 67 passing C# unit tests with 100% line coverage
- Diagnosed and systematically eliminated 84 compilation errors and 50+ production runtime failures by enforcing deterministic verification rules
- Delivered Smart 3D ecosystem enhancements by validating production behavior across 34 mutually exclusive production test cases, then identifying an additional application freeze defect

Software Engineering Contractor

May 2025 – Aug 2025

Tektonux LLC

Huntsville, AL

- Authored technical documentation for C# unit, end-to-end, and Storybook tests, enabling test adoption and efficient onboarding
- Supported automated test design and execution (Playwright) to validate defense system behavior against visual and functional requirements
- Tested terrain server functionality using GeoTIFF DEMs and Cesium, ensuring replicability across environments
- Implemented user interfaces in TypeScript, CSS, and HTML, ensuring accuracy across development stages

Test Engineer

Dec 2022

Independent Client (E-Commerce)

Dallas, TX

- Released a customer-facing production web system, validating readiness and enabling immediate deployment and revenue impact
- Partnered directly with the business owner, confirming requirements and ensuring delivered behavior matched business expectations
- Tested system behavior, usability, and visual correctness to ensure production readiness prior to launch

EDUCATION

Bachelor of Science in Computer Science

The University of Alabama in Huntsville

Huntsville, AL

- **Minor:** Mathematics | **Concentration:** Artificial Intelligence & Data Science
- **Summa Cum Laude**

Associate of Science in Computer Science

Calhoun Community College

Decatur, AL

- **Emphasis:** Business Information Technology
via CIS Certificates — Python, Java, C#, C++; MOS: Associate — Excel, PowerPoint, Word
- **Summa Cum Laude | Top Ten Student Award**

AREAS OF EXPERTISE

- **Leadership & Management:** Technical Leadership, Stakeholder Management, Change Management, Technical Communication, Cross-Functional Collaboration, Technical Documentation
- **AI & Automation:** AI-Driven Engineering, AI-Driven Test Generation, Engineering Automation, Test Automation, Deterministic Verification, Workflow Orchestration
- **Systems & Testing:** Verification & Validation (V&V), Requirements Traceability, Root Cause Analysis, Systems Testing, Test Case Design, End-to-End Testing, Integration Testing, Regression Testing, Unit Testing